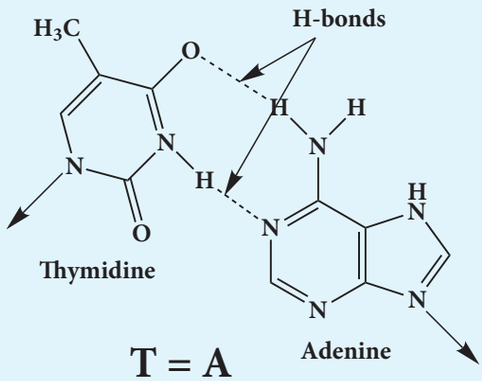
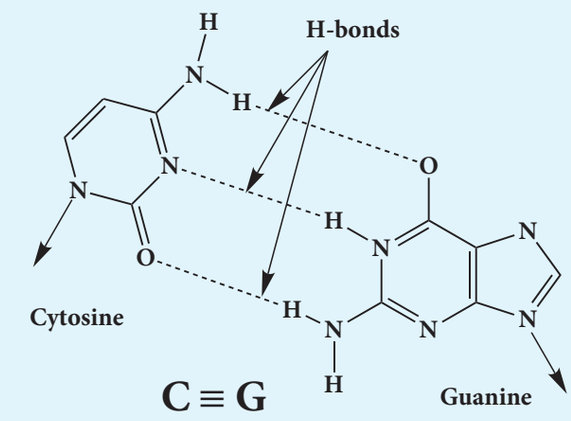
The structure elucidation of DNA by Watson and Crick in 1953 was a momentous event in science. They postulated a 3-dimensional model of DNA structure which consisted of two antiparallel helical DNA chains wound around the same axis to form a right-handed double helix.

The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water. The purine and pyrimidine bases of both strands are stacked inside the double helix, with their hydrophobic and ring structures very close together and perpendicular to the long axis, thereby reducing the repulsions between the charged phosphate groups. The offset pairing of the two strands creates a major groove and minor groove on the surface of the duplex.



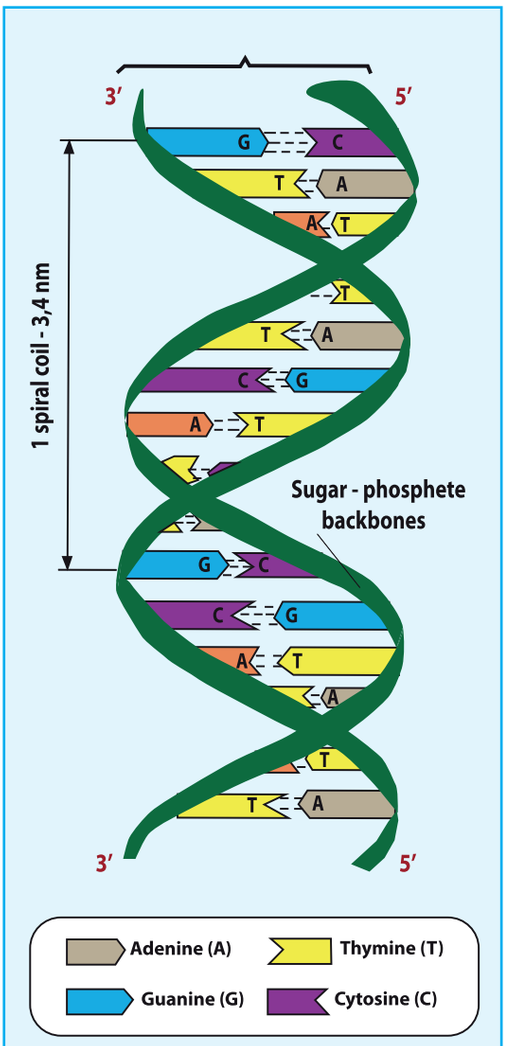




**Figure 14.21 DNA Double Helix**

The model revealed that, there are 10.5 base pairs (36 Å) per turn of the helix and 3.4 Å between the stacked bases. They also found that each base is hydrogen bonded to a base in opposite strand to form a planar base pair.

Two hydrogen bonds are formed between adenine and thymine and three hydrogen bonds are formed between guanine and cytosine. Other pairing tends to destabilize the double helical structure. This specific association of the two chains of the double helix is known



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